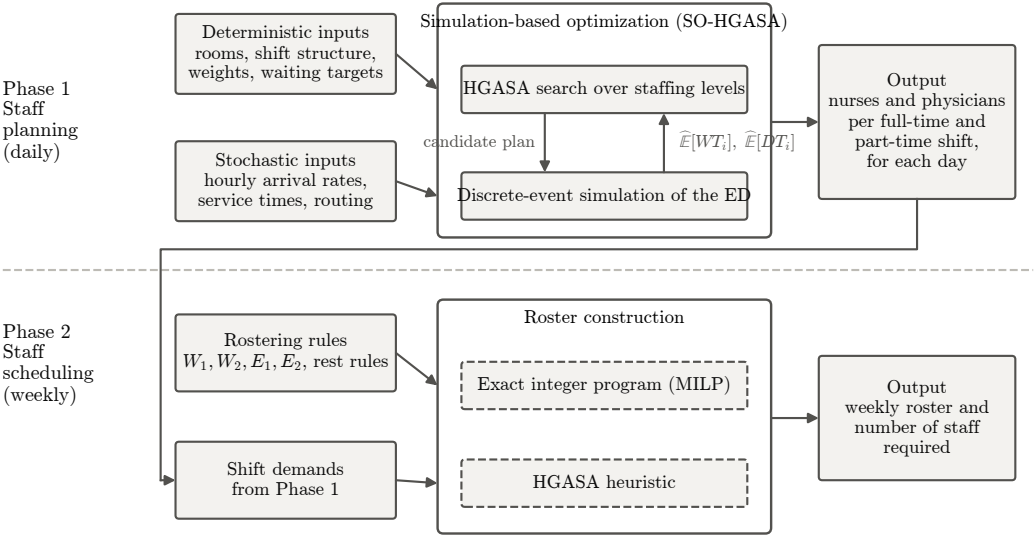
Phase 1
Staff
planning
(daily)
Deterministic inputs
rooms, shift structure,
weights, waiting targets
Stochastic inputs
hourly arrival rates,
service times, routing
Simulation-based optimization (SO-HGASA)
HGASA search over staffing levels
candidate plan
$\widehat{E}[WT_i]$, $\widehat{E}[DT_i]$
Discrete-event simulation of the ED
Output
nurses and physicians
per full-time and
part-time shift,
for each day
Phase 2
Staff
scheduling
(weekly)
Rostering rules
$W_1, W_2, E_1, E_2$, rest rules
Shift demands
from Phase 1
Roster construction
Exact integer program (MILP)
HGASA heuristic
Output
weekly roster and
number of staff
required